

How Should an Agentic RAG Assistant Hand Back a Decision That Depends on the User?

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Retrieval-augmented generation (RAG) assistants over internal documentation answer fluently even when the right answer depends on who is asking. Agentic RAG architectures prescribe a remedy: when the agent cannot settle a choice from the documents alone, it hands the decision back to the person. How that hand-back should look is untested where the agent’s recommendation can be wrong. We ran a 3×2 within-subjects study with $[N]$ university students who were new to their department’s GPU cluster. Each participant held a spec sheet for a course project and, on each of nine tasks, sent a scripted query, watched the assistant’s retrieval trace stream over a frozen documentation snapshot, and accepted or rejected the single answer it filled in. Every task was a fork the documents could not settle, and the deciding fact was on the sheet, which the assistant never saw. We varied only the form of the hand-back at the fork (modal stop, inline warning, or provenance-only citation), holding content, provenance and undo constant, and measured which sources were opened and for how long, whether participants kept or overrode the recommendation when it was right and wrong, decision time, and workload. *[Results and implications are added after data collection.]*

Additional Key Words and Phrases: agentic AI; retrieval-augmented generation; interruption; appropriate reliance; human-in-the-loop; proactive assistants

1 INTRODUCTION

Retrieval-augmented generation (RAG) is the usual way to put a large language model on top of an organization’s own documents, and internal documentation is full of answers that are conditional. On the Colorado State University (CSU) Computer Science department’s Falcon cluster, the quality-of-service (QoS) level a job is submitted under depends on how long the job will run: the documentation lists a `gpu_short` QoS with a 24-hour limit and a `gpu_medium` QoS with a 3-day limit, and which one applies depends on the job. How to connect depends on whether you are on campus; where a job should write depends on whether the data matters. People ask short questions that leave the deciding fact out (“Which QoS should I use to submit this job?”), and a RAG assistant composes a confident answer from whichever passage ranked higher. Nothing on screen says that the other passage applies to someone else, and a reader who is new to the system has no way to tell.

The reliance literature gives reason to expect that the reader will not catch it. Bansal et al. [1] found that explanations raise acceptance of AI recommendations whether or not they are right. Bućinca et al. [2] showed that cognitive forcing functions reduce overreliance but are the designs people like least. Vasconcelos et al. [13] modeled verification as a cost–benefit choice: people check the AI only when checking is cheap relative to the task. The practical question is therefore not whether to show sources but how to hand the decision back at the moment it is needed.

Agentic RAG architectures describe such a moment. Mishra et al. [8] catalogue a human-in-the-loop pattern in which the agent pauses execution to request disambiguation when it cannot resolve a choice itself. Sánchez-Adame and Mendoza [11] supply a contract, Minimal Proactivity Policies (MiPP), for how a system-initiated pause should be expressed, and tested three initiation forms in a within-subjects study of $n=48$, but on a system that was never wrong, so checking the source could not change any outcome. This paper asks what neither line has answered:

RQ. When an agentic RAG assistant finds mid-task that the right answer depends on a fact about the user that the question did not state, and hands the choice back, how does the form of that hand-back (modal stop, inline warning, or provenance-only citation) affect verification, appropriate reliance, decision time and workload?

We hold the agent constant and vary only the human-facing form. Each task is one scripted assistant session over a frozen snapshot of the Falcon documentation; the participant watches the retrieval trace, may open either of the two sources it cites, and accepts or rejects the one answer it fills in. Every task is a *context-dependent fork*: two passages give different answers, both correct for someone, and which applies depends on the participant’s situation, written on a spec sheet the assistant was never given. Such a fork is the case the human-in-the-loop pattern is written for: the agent cannot settle the choice from the documents and the person can, so pausing to ask is the pattern’s intended use, not a failure [4, 8]. The recommendation is wrong for this participant on about half of the tasks, so verifying has a payoff and keeping or overriding it can be scored as appropriate or inappropriate reliance [12].

This paper makes three contributions. (1) A comparison of three escalation forms at fixed decision points on an agent whose recommendation can be wrong, with verification and appropriate reliance as outcomes. (2) Predictions, derived from the interruption and reliance literatures, of how the forms trade verification against decision time and workload, and their test. (3) A reusable protocol (frozen snapshot, task records, contamination filter, matched task sets, event logging) that other groups can apply to their own documentation.

2 RELATED WORK

2.1 Interruption as a design variable

McFarlane and Latorella [7] made human interruption a first-class problem for interface design. McFarlane [6] compared four coordination methods (immediate, negotiated, mediated, scheduled) in a dual-task experiment with 36 participants and concluded that “negotiation support is the best overall solution except where small differences in the timeliness of handling interruptions is critical and then immediate is best.” Horvitz [4] framed mixed initiative decision-theoretically: initiate only when the expected utility of acting exceeds the cost of interrupting, and use dialogue to resolve uncertainty. Read against this work, and as our own interpretation, a modal stop is closest to McFarlane’s immediate method, an inline warning to a negotiated one, and a provenance-only answer does not interrupt at all. [verify] In these studies the interruption carried no fallible content; what they leave open is whether the coordination method also changes the accuracy of the decision the interruption asks the person to make.

2.2 Reliance on AI advice

Work at CHI, CSCW and IUI studies what people do with an AI recommendation that may be wrong. Bansal et al. [1] found no complementary gain from explanations, which raised acceptance regardless of correctness. Buçinca et al. [2] showed that forcing engagement before the AI’s answer is accepted reduces overreliance at a subjective cost. Vasconcelos et al. [13] showed across five studies that explanations help only when they lower the cost of verifying relative to the task. Schemmer et al. [12] define appropriateness of reliance in two dimensions: following the AI when it is right and overriding it when it is wrong. Möhle et al. [9] found in a judge–advisor study that a stated accuracy level shifted self-reported trust more when the reasoning was convincing, while actual accuracy was never varied. We take from this that self-report is a weak proxy for behavior, which is why our primary outcomes are logged. All of these

are single-shot decisions in which the AI proposes and the human disposes and none has a system-initiated pause mid-task, and none varies its form.

2.3 MiPP: a contract for proactive initiation

MiPP [11] separates decision authority (whether the system may initiate) from interaction authority (how the initiation is expressed). Its constraints include C1, a brief cue followed by a single question or choice; C2, a one-step escape hatch; and C3, an explicit provenance cue naming the source that justified the initiation. C5 says that when inputs are missing or ambiguous the system should “defer, downgrade to an indirect channel, or remain silent.” MiPP’s $n=48$ study compared a reactive baseline with three policy families (direct alert, confirmatory prompt, indirect entry) on productivity scenarios; the quieter families reduced perceived intrusion, and removing the confirmatory cue worsened intrusion, timing, trust and willingness. Three features of that study define the gap we address. Each condition bundled a family with its own trigger and scenario, so forms were not compared at the same decision points. Every initiation was correct by construction, so verification had no consequence and the families differed on it by at most six points. Its verification measure is anchored to an initiation, so it is undefined when nothing is announced. We anchor verification to the task. MiPP calls its contract domain-agnostic and names per-scenario designs with real workloads as future work.

C5 also creates a tension that our design tests. In MiPP, silence means non-initiation: when the inputs that describe the user’s situation are missing, the system does not initiate. Our agent is answering a query it was asked, so the move it can decline to initiate is the hand-back at the fork, and the fact it lacks is exactly the kind C5 names, a fact about the user. The provenance-only form is therefore non-initiation in C5’s sense: the assistant answers and cites but does not raise the fork. In MiPP’s setting that default is safe because nothing is lost by staying quiet; here it means quietly applying a recommendation that is wrong for this user.

2.4 Agentic RAG and RAG transparency

Mishra et al. [8] systematize agentic RAG and document a human-in-the-loop pattern that “models human oversight as a callable API within the action space,” pausing execution “to request disambiguation or supervision” when uncertainty exceeds a threshold. They note that existing implementations validate at predefined bottlenecks rather than triggering on internal ambiguity, and *propose* escalation precision and recall and detection of conflicting retrieved contexts as criteria for future systems; the paper is a survey, available only as an arXiv preprint, and evaluates nothing. Ravi and Sindhgatta [10] evaluated a RAG assistant for equity research with 50 financial professionals and report that source attribution and highlighting substantially improved self-reported trust and understanding, whereas confidence scores alone did not. Their measures are self-report, the system is not agentic, and nothing interrupts mid-task; a cited answer with named sources is the current practice that our provenance-only condition reproduces. Kirchenbauer et al. [5] validated FictionalQA items by prompting the generator blind and informed and discarding items answerable blind; the check reduces, not removes, the pretraining confound, and we reuse it as a filter on real Falcon facts.

2.5 Pre-generated agent behavior in reliance studies

Studies of reliance on AI advice typically fix the AI’s advice in advance and show every participant the same items, so that the correctness of each recommendation is controlled rather than left to a live model [2, 13]; the practice has the same aim as Wizard-of-Oz studies of intelligent interfaces, in which system behavior is controlled so that the interface

can be studied on its own [3]. We follow that paradigm: the agent’s behavior is generated once by the pipeline described in Section 3 and replayed identically to every participant.

2.6 The gap

Interruption research tells us how coordination method changes disruption when the interruption is not fallible. Reliance research tells us how forcing engagement changes overreliance in single-shot decisions. MiPP tells us how an initiation should be expressed, on a system that is never wrong. No study we are aware of has fixed the decision points of a fallible agent, varied only the form in which it hands a decision back mid-task, and measured whether people verify and rely appropriately, and at what cost.

3 METHOD

3.1 Participants

Twelve Colorado State University students who reported no prior use of the department’s Falcon cluster took part ([*n*] undergraduate, [*n*] graduate; [*n*] women, [*n*] men; courses of study were recorded at the session). They were recruited in person by word of mouth, were not compensated, and gave consent on the course’s class-project data-collection form. Three pilot participants run before data collection are not included.

3.2 Apparatus

Corpus. The corpus was a frozen snapshot of the CSU Computer Science department’s systems documentation (sna.cs.colostate.edu), crawled on 3 October 2026: 77 pages (23,965 words), stored as raw HTML and Markdown with per-page hashes, split at headings into 226 chunks and embedded with `nomic-embed-text`.

Drafting pipeline. The assistant’s retrieval and escalation decisions were produced before the study by a local pipeline (qwen2.5:7b-instruct through Ollama 0.35.1; temperature 0, seed 567) and frozen as task records, so no model ran during a session. [verify] Three elements of each task came from the pipeline: the retrieved passages and their ranking (an embedding search over the snapshot for the scripted query, top four chunks); the decision that the question was a fork, made by the fork check, which fired when the cosine margin between the top two chunks was at most 0.10 and an LLM judge found that the passages gave different correct answers depending on a fact the query did not state; and the recommendation, which was always the option whose passage ranked higher. [verify] The two options, the highlighted passage for each, the spec-sheet lines and the answer keys were written by us from the snapshot. The pipeline then had to find the same fork on its own: shown the retrieved passages, the judge named the options it saw, and a fork was used only when those options matched the two we had authored, which held for 8 of 25 query phrasings over 15 candidate forks. The seven-line trace shown to participants was a fixed template; the pipeline supplied what filled it: which page was Source A and which Source B (its retrieval ranking), that line 6 was a fork (its fork check), and which option line 6 recommended (the higher-ranked option). The wording of the lines themselves (“Reading the question...”, “Found Source A: ...”, “Source A suggests [option].”) was fixed, not written by the model, so that every task presented the same structure and the form of the fork line was the only variable.

Interface. [verify] The study ran as a single-page web application in a desktop browser: the block’s spec sheet, the task question, one answer field and Accept and Reject buttons on the left; the scripted query, a Send button and the assistant’s seven-line trace on the right, in which Source A and Source B appeared as chips that opened the frozen page

with the cited lines highlighted. Trace lines arrived one at a time at a fixed delay of 3–5 s stored in the task record, and only line 6, the fork line, was rendered differently across forms. Instrument screens between blocks and at the end of the session collected the questionnaires, and the application assigned each participant’s block order and task rotation from the participant number.

Logging. Every interaction (Send, each line onset, fork onset, modal pick, each source open and close with its duration, Reject, picker choice, Accept) was written to a local SQLite database under a participant code, together with a per-task summary of the final value, accuracy, following, verification, source pattern, dwell per source and decision time.

3.3 Tasks and Materials

The task pool held nine context-dependent forks, three per set (job submission; access and remote connection; storage and software), plus two practice tasks without a fork. A fork was two documented options whose correct choice depended on one fact the scripted query omitted and the spec sheet stated. Each fork existed in two twin versions with the same query, trace, sources and recommendation; the twins differed only in the sheet line and the task question, so that the recommendation was correct under one and wrong under the other. Each participant met every fork exactly once, in one of its two versions, one task from each set in every block, with the assignment of forks to forms and of twins to participants rotated so that every fork appeared under every form, and in each version, equally often. The cover story was a single course project on the cluster, and each block’s spec sheet listed the three deciding facts for that block’s tasks, one line per task. The recommendation was always the option whose passage ranked higher in retrieval, and the key was the option the documentation gave for the sheet line; correctness was never set by editing the agent.

3.4 Design

The experiment was a 3×2 within-subjects design with factors Escalation Form (S1 modal stop, S2 inline warning, S3 provenance-only) and Recommendation Correctness (correct, wrong). Each form was administered in one block of three tasks, one from each task set, so that task set was crossed with form rather than tied to it; the twin assignment was constrained so that every block contained at least one correct and one wrong recommendation and every participant received four or five of each. Block order was counterbalanced across the six permutations of the three forms, two participants per order, and the fork-to-form rotation was balanced within each order. With 12 participants this gave 108 measured tasks, 18 per Form $\times$ Correctness cell.

3.5 Procedure

[verify] Sessions lasted about one hour: instructions stating that the assistant was usually right but could be wrong and that any cited source could be opened; two practice tasks, during which the experimenter had participants open a source and use Reject and Accept once each; three blocks of three tasks, one block per form, each block’s three-line spec sheet staying on screen; the instrument screens after each block; and a closing ranking screen followed by a debrief. On each task the participant pressed Send, watched the trace, could open either source, and pressed Accept, or Reject followed by a choice from the two-option picker; no feedback on correctness was given.

3.6 Measures

Per task we recorded verification (a source viewed for at least 3 s of summed dwell before Accept), the source pattern under the same threshold and the raw dwell time per source, following (final value equal to the recommendation), accuracy (final value equal to the key) and decision time (fork-line onset to Accept). After each block an instrument screen collected raw NASA-TLX, a single perceived-intrusion item (“How intrusive was the assistant’s interruption?”, 1–7, the item used in MiPP-Eval, kept for comparability with MiPP), and a manipulation check asking which form the participant had just used (paused and asked, warned but continued, or cited only), and at the end of the session a screen collected a ranking of the three forms with a free-text reason.

3.7 Hypotheses and Analysis

We predicted that verification would be ordered $S1 > S2 > S3$ (H1), because designs that force engagement before an answer can be accepted raise engagement with the evidence [2] and an immediate interruption is handled most promptly [6]; that overriding a wrong recommendation would be higher under S1 and S2 than under S3 while following a correct one would not differ (H2, a Form $\times$ Correctness interaction), because forcing functions reduce overreliance [2] whereas a cited answer alone raises acceptance regardless of correctness [1], scored as relative self-reliance and relative AI reliance [12]; that decision time would be ordered $S1 > S2 > S3$ (H3a), because an immediate interruption costs the interrupted task the most [6] and checking a source is a time cost people weigh [13]; and that perceived workload would follow the same order (H3b), because forcing functions are rated as more effortful [2] and quieter initiations as less intrusive [11]. Task-level outcomes (H1, H2, H3a) were analyzed with logistic mixed models with fixed effects of form, correctness, their interaction and block position, and random intercepts for participant and task; decision time with a linear mixed model on log time; and the per-block scales (H3b) with repeated-measures ANOVA.

4 RESULTS

[To be written after data collection.]

4.1 Verification

4.2 Appropriate reliance

4.3 Decision time and workload

4.4 Subjective ranking and reasons

5 DISCUSSION

[To be written after data collection.]

5.1 Implications for proactive assistants

5.2 Limitations

5.3 Future work

6 CONCLUSION

[To be written after data collection.]

USE OF AI

Claude Code (Anthropic), running Claude Opus 5.5 with Claude Fable 5.1 subagents, was used in preparing the proposal that this paper grew from to: critique the instructor’s feedback and the original proposal; read the full text of the MiPP paper and check our claims against it; check each bibliographic entry against publisher records (Crossref, IEEE Xplore, Springer, ACM, ICLR proceedings, arXiv) and produce corrected BibTeX; draft the text of the proposal; and draw its mock-ups. For this checkpoint, Claude Code (Claude Fable 5.1) ported the abstract, introduction and related work into the review template, drafted the opening sentences of the Method section from the released apparatus code and data, and built the apparatus code described in Section 3 with the authors. The authors made every design decision, revise the text, and verify each reference on the publisher page before submission; claims the tool could not confirm are marked in the source and in the text.

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